

Chapter 3 - Practice Set

Operators

Instructions: Write each answer as its own separate Python file and run it. All problems are coding tasks -- write actual working Python code for each one.

Question 1 -- Arithmetic Calculator

Write a program that asks the user to enter two integers (using `input()` and type casting from Chapter 2). Then compute and print ALL seven arithmetic results: addition, subtraction, multiplication, true division, floor division, modulo, and exponentiation (first number raised to the power of the second). Label each result clearly in the output.

Question 2 -- Division Difference

Write a program that assigns the integer values 17 and 5 to two variables. Print the result of true division (`/`) and floor division (`//`) between them. Below each result, also print the data type of that result using `type()`. Observe and confirm that `/` always returns float and `//` returns int when both operands are integers.

Question 3 -- Even or Odd Checker

Write a program that asks the user to enter a whole number. Using the modulo operator (`%`), compute the remainder when the number is divided by 2. Print the remainder. Then, on the next line, print whether the number is even or odd by directly printing the result of the comparison (`number % 2 == 0`) as a boolean value.

Question 4 -- Comparison Table

Write a program that assigns the values 25 and 40 to two variables. Then print the result of all six comparison operators (`==`, `!=`, `>`, `<`, `>=`, `<=`) applied to these two variables, one per line, with a descriptive label for each (for example, 'Equal to: False').

Question 5 -- Logical Gate Simulator

Write a program that assigns boolean values to two variables p and q, then prints the result of: (p and q), (p or q), (not p), and (not p and q). Run your program four times by changing the values of p and q to cover all four combinations: (True, True), (True, False), (False, True), (False, False). Include all four versions in the same file using separate print statements with clear labels showing which values of p and q are being used.

Question 6 -- Score Tracker with Assignment Operators

Write a program that starts with a variable score = 0. Then simulate a simple game by applying the following updates in order, each using a compound assignment operator, and printing the score after each update: add 50, add 25, subtract 10, multiply by 2, use floor division by 3, raise to the power of 2 using `**=`. Print the score after every single step.

Question 7 -- Temperature Range Validator

Write a program that asks the user to enter a temperature value (as a float). Using a single chained comparison expression (do not use if/else), print the result of checking whether the temperature is between -10.0 and 100.0 inclusive. The output should be a single True or False boolean value. Then, using logical operators, also print the result of checking: the temperature is either below 0 or above 37, using or.

Question 8 -- Bitwise Explorer

Write a program that assigns the integer 52 to a variable p and the integer 14 to a variable q. Compute and print: the binary representation of p using `bin()`, the binary representation of q using `bin()`, the result of p & q and its binary form, the result of p | q and its binary form, the result of p ^ q and its binary form, the result of p << 1 and its binary form, the result of p >> 2 and its binary form. Label every single line of output clearly.

Question 9 -- Precedence Predictor

Write a program that prints the result of each of the following expressions WITHOUT changing anything -- write them exactly as shown and observe the output: (a) `3 + 4 * 2`, (b) `(3 + 4) * 2`, (c) `2 ** 3 ** 2`, (d) `(2 ** 3) ** 2`, (e) `10 - 6 // 2 + 1`, (f) `10 % 3 + 4 * 2 ** 2`. For each expression, write a comment in the code explaining the evaluation order you expect before running the program, then verify by running it.

Question 10 -- Mini Grade Classifier

Write a program that asks the user to enter a numeric exam score (0 to 100). Using ONLY arithmetic, comparison, and logical operators (no if/else -- just operator expressions), compute and print boolean results for each of the following: is the score a passing grade (greater than or equal to 50), is the score a distinction (greater than or equal to 75 AND less than or equal to 100), is the score outside the valid range (less than 0 OR greater than 100), is the score in the average range (greater than or equal to 50 AND less than 75). Print a labeled True/False result for each of the four checks.
